

KINE 1020 - Fitness and Health Laboratory Manual

Lab 2

2026-2027

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Frequently Asked Questions

- **Lab Manual:** The lab manual will be posted in twelve parts on eclass (one for each lab). Make sure that you have an online or printed version of the lab manual available for you to refer to during the lab.
- **Worksheets:** You should print the worksheets for each lab prior to attending the lab each week. Alternatively, you can complete online versions of the worksheet, but due to the nature of the labs, it is easier to have printed copies, but this is NOT mandatory
- **Attendance:** Laboratories are in person (Room: **HNES 018**) and it is mandatory for you to attend. Labs occur ****BIWEEKLY**** (i.e., you attend your IN-PERSON labs every other week in the timeslot indicated on your official timetable). They generally are around ~1 hour and 50 min in length. There is **NO** virtual Laboratory option. Please come prepared with your lab manual each time.
- **Lab Section:** You will work with and learn from the same teaching assistant (TA) for your labs. Please check your York University provided schedule/timetable for your 2-hour lab section. It will be in a 2-hr slot on a weekday (Monday to Thursday) between 8:30AM and 8:30PM.
 - You must **only attend your** scheduled laboratory. Permission may be granted by the TA ahead of time for you to attend another lab but only under extenuating circumstances, and on a case-by-case basis. If you show up to another lab without acquiring permission from that TA first, you will be turned away.
 - If you need to change lab section for any reason, this must be formally changed by contacting either the Kinesiology office (ugkhs@yorku.ca) or the registrar's office. Again, you **cannot** just show up to a different lab time.
- **eClass:** Your eClass site will only allow you access to the lab section that you are registered for. If you do not see your lab section on your eClass, please let us know *via* the Kine 1020 email (kine1020@gmail.com). Please learn your TA's name as it will be needed for your midterm and final exam. Their name and email address will be posted under the Laboratory section on the eClass web page.
- **Grading:** You will be graded equally on your participation in each laboratory and the quality of the work you hand in. Together each laboratory is valued at **2% of your final grade** (1% for laboratory attendance and 1% for completed worksheet). You can miss two laboratories for personal reasons, without penalty, and submit up to 1 accommodated lab assignment for an additional missed lab unless there are extenuating circumstances. Only your top 10 grades from the 12 total labs will be counted for a maximum of 20% of your final grade.
 - **Accommodated Lab Hand-Ins:** Students with *approved accommodations* for in-person attendance, can submit an accommodated lab. This entails a written summary of the lab along with the relevant completed worksheets to earn 2% for that laboratory. Approved students will receive a dataset for the case-studies from their TA so that they can complete the worksheets on their own. Unless there are extenuating circumstances, the summaries and completed worksheets must be handed in on eClass within 1 week of the completion of your normally scheduled lab. In cases where that is not possible, such as sudden and/or prolonged illness, students can discuss specific accommodations needed with their TA on a case-by-case basis.
 - **Hand-In:** You should hand in your laboratory assignment/report in lab to your TA at the end of the lab. If you need more time to complete the lab, you must hand in your assignment/report *within 24 hours* from the END of your respective laboratory *via* eClass. ALL submitted work for KINE 1020 must include your full name, student number, lab group number and TA's name. Please submit the file as a PDF or word document (note: as a York University you can download the Microsoft office suite for free)

- **Clothing/Attire:** For all labs, please dress in appropriate clothing (e.g., athletic clothing and shoes, comfortable to move in clothing). In the winter, you must bring clean/dry shoes to avoid creating a slipping hazard from snow/rain.
- **Food & Drink:** Food and drink is not permitted within the lab, with the exception of a water bottle. Please notify your TA if other food/drink is required for medical purposes.

Common Unit Conversions

Length

1 m	=	100 cm
1 inch	=	2.54 cm
1 ft	=	12 inches
1 mile	=	1.6 km

Weight

1 kg	=	1000 g
1 kg	=	2.2 lb

Time

1 hour	=	60 minutes
1 min	=	60 seconds

Energy

1 L O ₂	=	5 kcal
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Laboratory #2 – Stress Management and Relaxation

The material in this laboratory is adapted with permission from the copyrighted 2004 CD ROM Owner's Manual for Self-Regulation of Your Brain and Body, V.E. Wilson & M.S. Cummings, YSAM Inc., Toronto.

"Balance" is a basic principle that applies to every aspect of your life. The body/mind has built-in balancing mechanisms to keep the body operating efficiently without us having to pay attention. Cannon (1963) called this ability of the body/mind to be aware of its own state and operate efficiently; "the wisdom of the body". Unfortunately, it seems that the pace of life is accelerating faster every day, making it even tougher to avoid losing balance in many areas of our lives. Being out of balance creates stress and makes us more reactive so that new stressors can push us even further out of balance (Weil, 1974) (i.e., the ripple effect). Most of the common health and performance problems facing society today can be traced back to lifestyle imbalances. Some of our ways of coping with these problems are ineffective and actually make these problems worse (i.e., excessive drinking, overeating, etc.).

Chronic stress levels may increase due to significant life events and life transitions (such as death of a loved one, starting university and/or being in a global pandemic), but more often it is a cumulative effect of several little daily hassles that were not effectively coped with. Sometimes the stress that you think you can handle and what your body can *actually* handle can be quite different. Every system in the body, including the mind, will break down if you chronically stress it enough. Chronic muscle tension due to stress wastes significant amounts of your available energy and is comparable to leaving one foot on the brake as you press the gas pedal. Chronically stressed muscles can lead to headaches and back aches, and chronic worry can lead to depression. Suppressed emotions can contribute to disease and poor interpersonal relationships (Girdano, Dusek, & Everly, 2008).

One of the biggest challenges in stress management is that individuals may be desensitized to their body's signals regarding their actual chronic stress levels throughout the day. It appears that if you stay in a bad situation long enough, you just get used to it (Girdano, Dusek & Everly, 2008) and your chronic stress will be re-set as the new 'normal' if you do not have a healthy reference point for comparison.

It is hard to do something about a problem if you are not aware that you have one.

Through practice of relaxation techniques, you will become more aware and connected with your own bodily sensations and will be better able to detect differences in tension and relaxation in the different areas of your body. Another way to enhance awareness of our tension levels is to measure our physiology through biofeedback measures such as peripheral skin temperature. Peripheral skin temperature gives us an indirect measure of the autonomic nervous system tension. As part of the normal stress response, blood flow tends to be shunted away from the extremities, resulting in a lower peripheral skin temperature. As you relax, your peripheral skin temperature will rise due to increased blood flow returning to the extremities. With relaxation training, one should try to achieve peripheral skin temperature goals of about 35°C/95 °F through relaxing their autonomic nervous system. (Wilson & Somers, 2011).

In this laboratory we will complete a brief self-assessment of your stress level, assess some objective measures of stress, and practice some relaxation techniques. The first step is for you to first become aware of what your typical reactions are to stressful events. The effects of stress may manifest in different ways such as in your brain, body and behaviour. It is normal and healthy to have some signs of stress, but it becomes unhealthy and unproductive to have too many signs of stress or extreme stress reactions (such

as migraine headaches, irritable bowel syndrome, panic attacks, etc.). It is also unhealthy to have only a few signs of stress but have them consistently for a long time period.

When you recognize that you are under stress, it is important to be aware of some basic techniques that can help bring us back into balance. In this lab, we will learn deep rhythmical healthy breathing, and use breathing to activate or relax the brain and body.

CAUTION: Self-regulation can be used to prevent or lessen signs of stress or illness. It is possible that frequent or severe stress signs may indicate medical problems that require medical attention from your physician.

Some conditions need special attention when learning self-regulation. Individuals who have epilepsy, diabetes, low or high blood pressure, muscle disorders, breathing disorders, or are receiving professional psychological counseling should seek medical advice prior to beginning a program of self-regulation training. Some medications such as muscle relaxants, tranquilizers, insulin, or others affected by changes in metabolism need to be monitored in case dosages need to be changed.

Breathing

Breathing is a basic rhythm in the body that, if in an unhealthy pattern, can affect everything else. It is easily integrated since you have to breathe in almost all situations. By simply adjusting the pattern of your breathing, you can use it to get yourself more revved up (emphasizing the inhalation) or slowed down (emphasizing the exhalation/pause phase).

Improper breathing may lead to:

- Difficulty talking or moving normally
- Forgetfulness
- Becoming flustered or anxious (anxiety/panic attacks)
- Headaches or excessive muscle tension
- Racing heartbeat or racing thoughts

Methods

Breathing Assessment Are You a Good Breather?

Circle the number on the scales below that seems to be the most accurate assessment of your breathing at this moment.

1. **SPEED:** Count the number of breaths that you take in one minute while at rest with no physical activity. One breath is from the time you begin to inhale until the next time you begin to inhale on your next breath. The average breathing rate is 12 to 14 breaths per minute.

18	15	12	9	6
Fast		Average		Slow

2. **RATIO:** Without changing your breathing rate, time the length of your inhalation (in seconds). Then time the length of your exhalation (from when you begin your exhalation to when the next inhalation begins). Do this for 2 or 3 cycles.

-2	-1	0	+1	+2
Longer Inhalation		Even		Longer Exhalation

Typical breathing patterns will have an equal inhalation and exhalation. This typical breathing pattern has a shorter than ideal exhalation phase. This is because individuals believe they need to begin to inhale earlier than they need to (habit air hunger). If they inhale even earlier, this will lead to hyperventilation (psychological air hunger). The point of these exercises is to lengthen the exhalation and inhale at the point of true physiological air hunger. Relaxed breathing patterns will have a longer exhalation than inhalation, while an aroused state will have a longer inhalation than exhalation. To relax it is easiest to focus on lengthening your exhalation. However, trying hard to relax can actually cause you to become tenser. Think about what happens if you really try hard to hit a golf ball or to get to sleep. You often get the opposite response by hitting the shot poorly or lying there wide-awake. It is important that you approach the self-regulation exercises with a passive or “try easy” attitude. Like any motor skill, practice with feedback is necessary. After four weeks of practice, most individuals report significant changes in their awareness and control of their brain, body and behaviour. *With self-regulation techniques, you need to practice the skills when you don't need them so that they are there when you do need them.*

Peripheral skin temperature can range from 8 to 37°C, and the goal with the exercises in this laboratory is to bring your peripheral skin temperature to 35°C. Your degree of relaxation is influenced by genetics, and is reduced by fasting, smoking, low external air temperature, and stress and is increased by exercise.

CONTROL OF THE DEGREE OF RELAXATION/ENERGIZATION

You are always in control of how relaxed or energized you wish to feel. If you want to stop feeling deeply relaxed and become more alert and energized during a relaxation exercise:

1. Take a deep breath
2. Open your eyes
3. Bend arms quickly 3-4 times
4. Blink your eyes
5. Say or think an energy command like "let's go!" or “do it!”.

Methods

For this laboratory, we will examine signs of stressors and measure changes in peripheral skin temperature with two different breathing techniques.

- Complete the first two questions of the lab identifying signs that you are stressed.
- You will then be advised to take a mat and sit in your own space on the floor in the lab. Please note, there is a small chance of fainting when performing the breathing techniques.
- Attach the wire of the skin thermometer to the bottom of the first knuckle of the middle finger with masking tape. It is not necessary to completely unravel the wire.

- Make sure that your hand is in a relaxed position with your palm facing up.
- For both breathing exercises, sit relaxed on a mat.
- Close your eyes
- Roll your head around until your head feels balanced on your neck
- Bring your head back slightly and then bring your head forward till your neck muscles don't need to work (balanced).
- Perform Technique #1.
- Repeat and perform Technique #2.
- Be careful not to break the wire when removing the tape at the end of the lab.

Technique #1 – AHHSOME© (2x's for 10-20 sec.)

Several systems need to relax quickly in order to facilitate a relaxation response without interfering with your normal daily routine. AHHSOME releases tension in key muscle groups, encourages effective breathing, enhances good blood flow and moves your attention back into the present. This is a favourite technique for many athletes as it allows them to move into the "try easy" or the 80% effort to the 100% performance state (Wilson & Cummings, 2011).

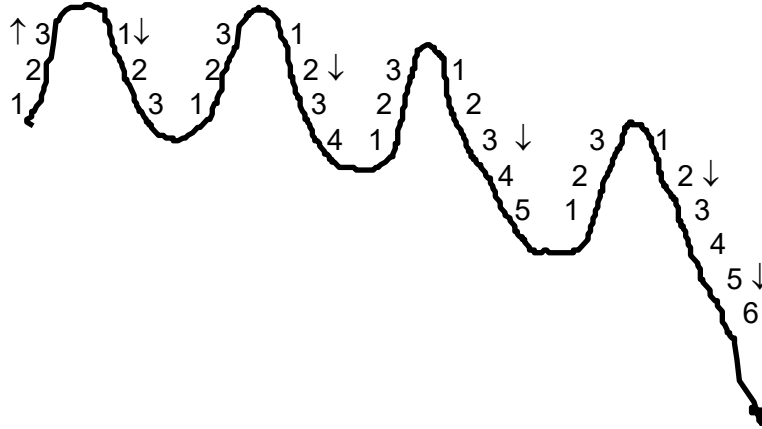
- Take in an easy deep breath, as you exhale slowly, feel and let go of the muscle tension in your forehead, eyes, jaw, shoulders, elbows and hands.
- As you take a second easy deep breath, slowly exhale: drop your jaw and shoulders. Imagine opening up the blood vessels carrying oxygen rich blood from your heart and let the blood flow freely down to your hands. Allow awareness of body sensations to primarily rest in the hands and face by the end of this breath.
- Complete this exercise by refocusing your attention by using an action word or question such as "What do I want here now?" or "What is it that I have to do here to get what I want?"

Technique #2 – QUANTITATIVE BREATHING (5 minutes)

Extended exhalation/pause phase. Make the number of counts as you exhale out a little longer than the counts as you inhale. For example, if it takes three counts to get the air in, then make it 4 counts to get the air out. As you practice and become more skilled, attempt to make the number of counts out twice as many as the number in. All the extra time spent in the exhalation/pause phase creates greater relaxation with each breath. Other points to make this breathing technique most effective are:

- Inhale through your nose evenly (do not breathe through your mouth)
- At the top of the inhalation, relax or let go of whatever muscle tension you used to get the air into your body
- Exhale through your mouth letting the jaw and shoulders drop (purge your excess energy)
- Let the air flow out of the body by itself and pause in the quiet spot so that you spend a little longer on the exhalation/pause phase with each breath until you are spending twice as long on this phase compared to the inhalation phase. The ratio of the inhalation and exhalation is more important than the absolute time of each phase.
- It may be helpful to slowly progress from a 3-3 to a 3-6 inhalation/exhalation ratio as illustrated by the figure below (i.e., breathe in – 1,2,3; breathe out – 1,2,3; breathe in – 1,2,3; breathe out – 1,2,3,4, breathe in – 1,2,3, breathe out – 1,2,3,4,5, etc.).
- Attempt to make this breathing action as smooth as possible.

Relax



- For people with busy minds, counting silently to yourself and seeing the numbers can help you relax better.
- Try this now but first record your peripheral skin temperature reading and notice how your mind and body feels.
- After 5 minutes of practice, notice how your mind and body feel and record your post peripheral skin temperature reading and any mind/body changes you experienced.
- Make sure that you do not do more than 5 minutes as this may bring about other negative effects or hyperventilation.

These are a few basic techniques that are the foundation of self-regulation. If you are interested, more advanced techniques will be covered in KINE 4420 Relaxation: Theory and Practice.

Lab 2 Worksheet

What are the three signs that let you know that you are under significant stress?

1. _____
2. _____
3. _____

Check and see how many stress signs you usually experience?

Brain

- | | |
|--|---|
| ☐ Can't concentrate | ☐ Can't remember |
| ☐ Easily confused or distracted | ☐ Frequent mood swings |
| ☐ Easily overwhelmed | |

Body

- | | |
|--|---|
| ☐ Headaches | ☐ Upset stomach |
| ☐ Muscle aches/pains | ☐ Teeth grinding/RSI |
| ☐ Immune Dysfunction (colds, flu, etc.) | |

Behaviour

- | | |
|---|---|
| ☐ Short fuse | ☐ Poor performance |
| ☐ Can't sleep | ☐ Poor Diet |
| ☐ Increased substance use | |
| ☐ Hardening of the attitudes (My way or the highway) | |
| ☐ Become socially isolated or emotionally distant | |

Assessment of Relaxation (Pre/Post Skin Temperature)

Technique	Pre skin temperature	Post skin temperature	List any subjective body/mind changes that you felt after practicing these breathing techniques (e.g., warmer, cooler, muscles less tense, more alert, less alert, etc.)
Ahhsome			
Quantitative Breathing			

Hand-In Pages

Hand-In Sheets

Please complete the following sheets and hand them to your lab instructor at the end of each lab.

Some of the hand-in sheets are redundant with the worksheets for the lab. This is on purpose so that you will have a copy of your results and work so that you can study for the exam. You will need to hand in the hand-in sheets in order for you to receive a grade for each laboratory.

Lab 2 Hand-In Data Sheet

Name: _____

Student Number _____ Section: _____

What are the three signs that let you know that you are under significant stress?

1. _____
2. _____
3. _____

Check and see how many stress signs you usually experience?

Brain

- | | |
|--|---|
| ☐ Can't concentrate | ☐ Can't remember |
| ☐ Easily confused or distracted | ☐ Frequent mood swings |
| ☐ Easily overwhelmed | |

Body

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| ☐ Headaches | ☐ Upset stomach |
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| ☐ Immune Dysfunction (colds, flu, etc.) | |

Behaviour

- | | |
|---|---|
| ☐ Short fuse | ☐ Poor performance |
| ☐ Can't sleep | ☐ Poor Diet |
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Technique	Pre skin temperature	Post skin temperature	List any subjective body/mind changes that you felt after practicing these breathing techniques (e.g., warmer, cooler, muscles less tense, more alert, less alert, etc.)
Ahhsome			
Quantitative Breathing			